

Aim:

To design single page application using React JS.

Description:

The webpage is developed using **React.js**, a JavaScript library for building user interfaces. The application consists of multiple components such as Login and CRUD, created using functional components and React Hooks.

React concepts and features are used:

- **React Functional Components** – To create reusable UI sections (Login and CRUD pages).
- **useState Hook** – To manage state such as ID, password, form inputs, and CRUD data.
- **React Router** – To navigate between Login and CRUD pages.
- **Event Handling** – To handle form submission and button click actions.
- **Conditional Rendering & Validation** – To validate login credentials and display error messages.
- **CRUD Operations** – To perform Create, Read, Update, and Delete operations dynamically.
- **CSS Styling** – To design and enhance the user interface with proper layout and color combinations.

Code:**App.jsx:**

```
import { Routes, Route } from "react-router-dom";
import Login from "./Components/Login";
import Crud from "./Components/Crud";
import "./App.css";
function App() {
  return (
    <Routes>
      <Route path="/" element={ <Login /> } />
      <Route path="/crud" element={ <Crud /> } />
    </Routes>
  );
}
export default App;
```

Login.jsx:

```
import { useState } from "react";
import { useNavigate } from "react-router-dom";
export default function Login() {
  const [id, setId] = useState("");
  const [pass, setPass] = useState("");
  const [error, setError] = useState("");
  const navigate = useNavigate();
  const handleSubmit = (e) => {
    e.preventDefault();
```

```

if (id === "" || pass === "") {
  setError("Both fields required!");
}
else if (id === "admin" && pass === "admin") {
  setError("");
  navigate("/crud");
}
else {
  setError("Invalid ID or Password!");
}
};

return (
  <
    <h2 className="student-name">Ajay Kumar - 2024503701</h2>
    <div className="login-container">
      <div className="login-box">
        <h1>Login</h1>
        <form onSubmit={handleSubmit}>
          <input
            type="text"
            placeholder="Enter ID"
            value={id}
            onChange={(e) => setId(e.target.value)}
          />
          <input
            type="password"
            placeholder="Enter Password"
            value={pass}
            onChange={(e) => setPass(e.target.value)}
          />
          <button type="submit">Login</button>
        </form>
        <p className="error">{error}</p>
      </div>
    </div>
  </>
);
}

```

Crud.jsx:

```

import { useState } from "react";
export default function Crud() {
  const [name, setName] = useState("");
  const [data, setData] = useState(["playing"]);
  const [editIndex, setEditIndex] = useState(null);

```

```

const handleAdd = () => {
  if (name === "") return;
  if (editIndex !== null) {
    const updated = [...data];
    updated[editIndex] = name;
    setData(updated);
    setEditIndex(null);
  } else {
    setData([...data, name]);
  }
  setName("");
};

const handleDelete = (index) => {
  setData(data.filter((_, i) => i !== index));
};

const handleEdit = (index) => {
  setName(data[index]);
  setEditIndex(index);
};

return (
  <div className="crud-container">
    <div className="crud-box">
      <h2>CRUD Operations</h2>

      <div className="crud-input">
        <input
          type="text"
          placeholder="Enter Name"
          value={name}
          onChange={(e) => setName(e.target.value)}
        />
        <button onClick={handleAdd}>
          {editIndex !== null ? "Update" : "Add"}
        </button>
      </div>
      <ul>
        {data.map((item, index) => (
          <li key={index}>
            {item}
            <div>
              <button onClick={() => handleEdit(index)} >Edit</button>
              <button onClick={() => handleDelete(index)} >Delete</button>
            </div>
          </li>
        ))}
      </ul>
    </div>
  </div>
)

```

```
    )})  
</ul>  
</div>  
</div>  
);  
}
```

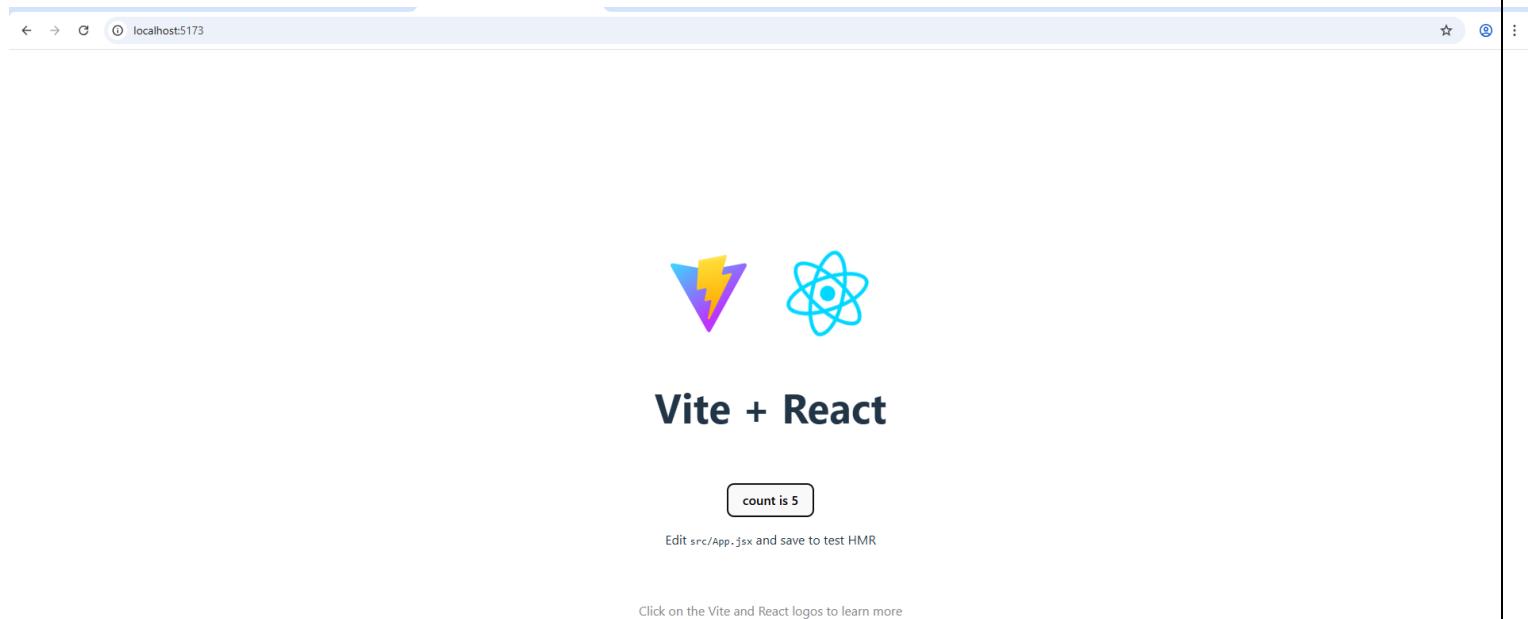
App.css:

```
body{  
  background-color: lavender;  
}  
.login-container {  
  height: 100vh;  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  background-color: lavender;  
}  
.login-box {  
  background: white;  
  padding: 30px;  
  width: 300px;  
  border-radius: 10px;  
  box-shadow: 0px 4px 15px rgba(0, 0, 0, 0.1);  
  text-align: center;  
}  
.login-box input {  
  width: 100%;  
  padding: 10px;  
  margin: 10px 0;  
  border: 1px solid #ccc;  
  border-radius: 5px;  
}  
.login-box button {  
  width: 100%;  
  padding: 10px;  
  background-color: #4caf50;  
  color: white;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
}  
.login-box button:hover {  
  background-color: #45a049;  
}  
.error {  
  color: red;
```

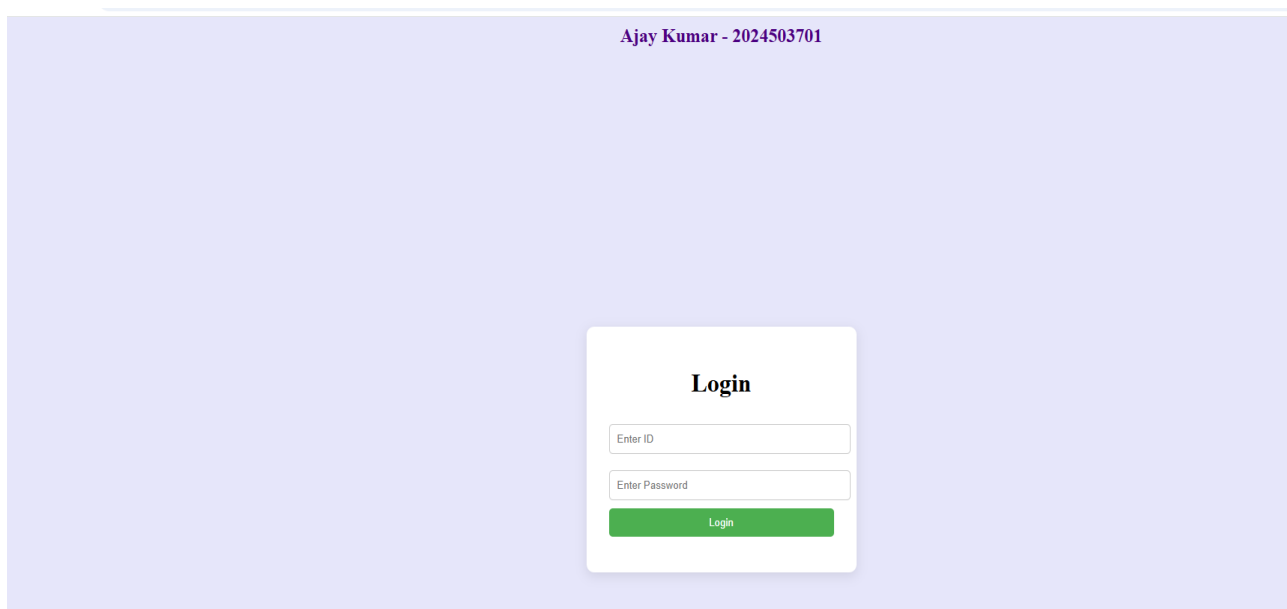
```
margin-top: 10px;
font-size: 14px;
}
.crud-container {
display: flex;
justify-content: center;
margin-top: 60px;
}
.crud-box {
width: 400px;
}
.crud-input {
display: flex;
gap: 10px;
margin-bottom: 20px;
}
.crud-input input {
flex: 1;
padding: 8px;
}
ul {
list-style: none;
padding: 0;
}
li {
display: flex;
justify-content: space-between;
padding: 8px;
border-bottom: 1px solid #ddd;
}
button {
padding: 6px 12px;
cursor: pointer;
color: #4b0082;
background-color: #e6e6fa;
border: 1px solid #d8bfd8;
border-radius: 6px;
transition: 0.3s ease;
}
button:hover {
background-color: #d8bfd8;
color: white;
}
.student-name {
```

```
text-align: center;  
margin-top: 10px;  
color: #4b0082;  
font-weight: bold;  
}
```

Output:



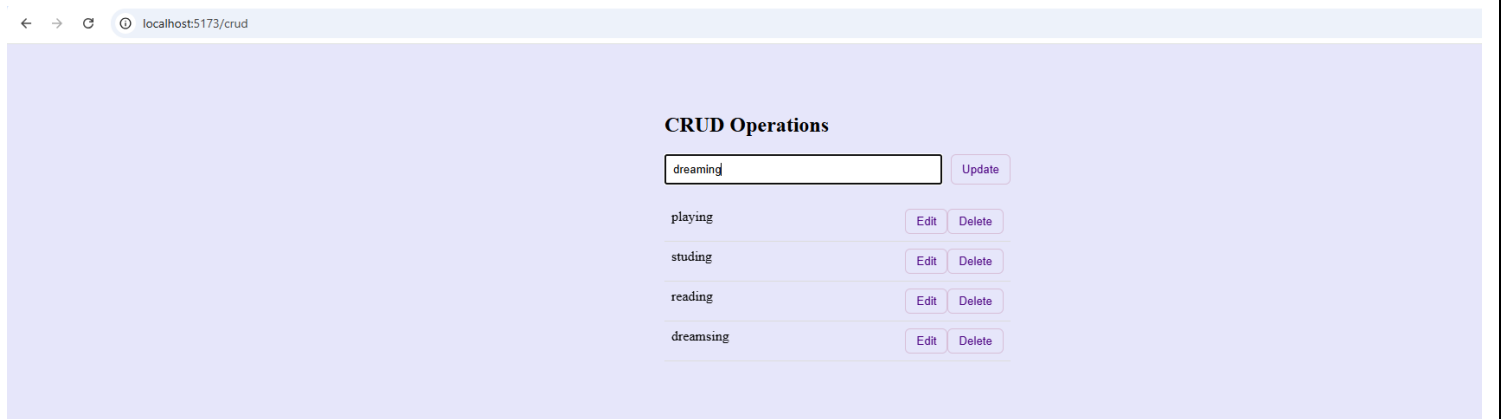
Figure(a) Localhost by Default



Figure(b): Login Page



Figure(c): Addition on ToDo



Figure(d): Updation on ToDo



Figure(e): Deletion on ToDo

Result:

Thus, a functional and interactive web application was developed using React.js and CSS. It includes a login page with validation and a CRUD module to perform Create, Read, Update, and Delete operations. The project demonstrates component-based design and state management using React Hooks.